

PaperMythXR: Embodied Mixed-Reality Storytelling via Chinese Paper-Cutting

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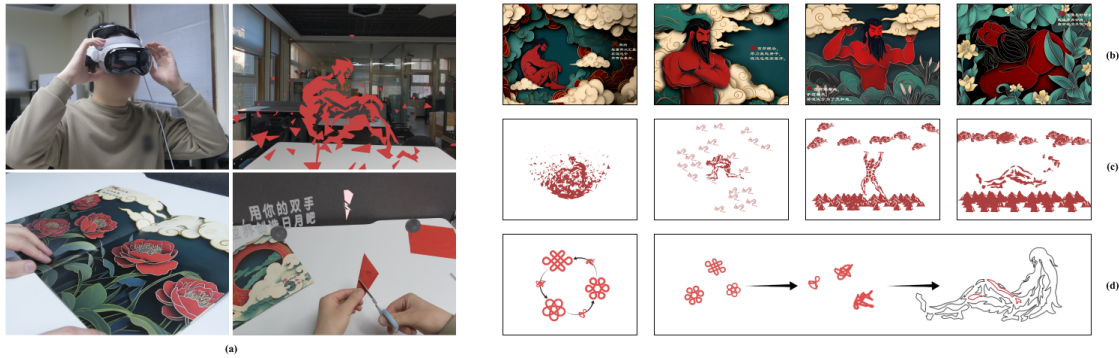


Figure 1: Overview of PaperMythXR, illustrating how the system fuses physical and digital worlds. (a) Mixed-reality interaction: user reading, cutting, and viewing with Apple Vision Pro. (b) Visual design of the physical storybook. (c) Examples of narrative-driven virtual animations. (d) Transformation of the core Pan Chang motif from cut pattern to dynamic digital asset.

Abstract

Traditional Chinese paper-cutting embodies tactile skill and cultural heritage, yet most AR/VR systems separate physical making from narrative progression. PaperMythXR is a mixed reality (MR) system where real paper-cutting directly drives a mythological storyline—The Creation of the World by Pangu. The system unites a physical storybook, crafting tools, and a head-mounted MR display in a Read–Create–Advance loop, combining tangible creation with immersive digital scenes. A human–AI collaborative pipeline ensures consistent aesthetics between printed illustrations and

animated MR elements. In a two-stage public study (N=53), over 90% of participants reported higher immersion, stronger cultural connection, and greater creative agency. This work demonstrates how MR can fuse craftsmanship and storytelling, offering a design framework for cultural heritage and interactive education.

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1 Introduction

Traditional Chinese paper-cutting combines three inseparable values: visual artistry, manual craftsmanship, and cultural narrative. Yet existing digital methods rarely preserve all three at once. VR simulations can mimic the process but lose the tactile feedback of real materials [Nie and Guo 2024]. AR overlays can animate finished

works but often break the causal link between physical making and narrative flow [Huang et al. 2023]. PaperMythXR addresses this gap by treating physical creation itself as the narrative engine. Inspired by prior work on using tangible artifacts as narrative triggers [Daffara et al. 2020], the system adopts a “Read–Create–Advance” loop, where users progress through a mythological story by reading a physical book, performing guided paper-cutting tasks, and watching their creations animate within the MR scene. Unlike prior systems that use crafted artifacts only as props [Song et al. 2020], our approach turns making into a causal driver, positioning the user as a co-author of the unfolding narrative.

2 Method

We implemented the “Read–Create–Advance” model through the Chinese creation myth Pangu Creates the World (see Fig. 2). The experience runs on Apple Vision Pro and centers on a continuous interaction loop. While reading a physical storybook, page recognition triggers corresponding MR scenes. At key narrative moments (e.g., Pangu forming the sun and moon), the system pauses for a hands-on task. The user follows an in-headset tutorial to complete a paper-cutting exercise, then places the finished cutout onto the storybook. The headset’s passthrough cameras capture this artifact in real time and trigger corresponding animation in the MR scene—for instance, a user’s cut sun rises into the virtual sky. By co-locating reading, making, and viewing within one workspace, the system maintains immersion and supports fine-motor crafting without breaking flow.

The technical pipeline ensures coherence between physical and virtual domains. A human–AI collaborative workflow was used for asset creation: diffusion models generated variations of the core Pan Chang motif (see Fig. 1), which were refined by an artist into illustrations and 3D animation assets. To achieve seamless transformations from print to animation and maintain high frame rates, we built a pipeline from Cinema 4D to Unity using vertex animation textures (VAT) for offloading computational burden to the GPU. This approach preserves cultural authenticity while enabling stable playback of complex, variable-topology animations in MR.

3 Results

We conducted a two-stage public study (N=53). In Stage 1 (N=43), participants experienced the passive MR narrative; 88% found transitions between book and animation smooth, and immersion ratings averaged 4.6/5. In Stage 2 (N=10), participants completed the full loop with paper-cutting tasks. Results were highly positive: cultural connection averaged 4.7/5 and creative agency 4.8/5. Many described the experience as “pushing the story forward like clearing a game level.”

Challenges included occasional confusion during scene transitions, lighting-sensitive recognition errors, and minor ergonomic strain during prolonged cutting with the headset. Despite these issues, the prototype ran stably across all sessions, validating the feasibility of creation-driven mixed reality storytelling.

Acknowledgments

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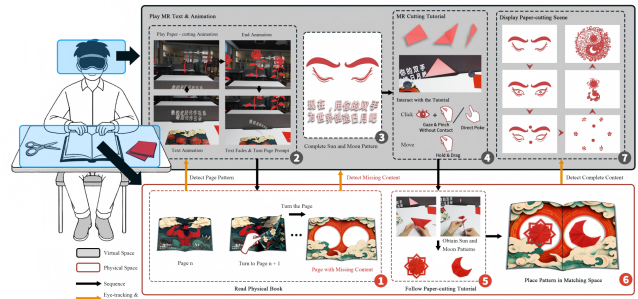


Figure 2: The Read–Create–Advance interaction loop of PaperMythXR. The system integrates physical and virtual spaces through three core components: (A) **Progressive Narrative Structure** — turning the physical storybook pages triggers MR text and animations (1–2); (B) **Interactive Crafting Mechanism** — guided paper-cutting tutorials and placement of physical cutouts (3–6); (C) **Dynamic Presentation Style** — user-created patterns transform into animated story elements (7). Black borders denote virtual spaces, and red borders denote physical spaces.

References

- Stephanie Claudino Daffara, Anna Brewer, Balasaravanan Thoravi Kumaravel, and Bjoern Hartmann. 2020. Living Paper: Authoring AR Narratives Across Digital and Tangible Media. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems (CHI EA '20)*. Association for Computing Machinery, New York, NY, USA, 1–10. doi:10.1145/3334480.3383091
- Yuchen Huang, Yulei Xu, Jiangping Yuan, Chaoyuan Zan, and Kaida Xiao. 2023. Research on the interactive perception of color paper-cutting culture rendering based on AR. *Journal of the Optical Society of America A* 40 (2023), 2059–2067. doi:10.1364/JOSAA.491545
- Kexin Nie and Mengyao Guo. 2024. Flying Your Imagination: Integrating AI in VR for Kite Heritage. In *SIGGRAPH Asia 2024 Posters (SA '24)*. Association for Computing Machinery, New York, NY, USA, Article 30, 2 pages. doi:10.1145/3681756.3697907
- Yingjie Song, Chenglei Yang, Wei Gai, Yulong Bian, and Juan Liu. 2020. A new storytelling genre: combining handicraft elements and storytelling via mixed reality technology. *The Visual Computer* 36, 10–12 (oct 2020), 2079–2090. doi:10.1007/s00371-020-01924-3